

Star chromatic index of complete graphs

OPG-37275

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Conjecture Is it possible to color edges of the complete graph K_n using $O(n)$ colors, so that the coloring is proper and no 4-cycle and no 4-edge path is using only two colors?

Equivalently: is the star chromatic index of K_n linear in n ?